

# Paper Replication Report #133: Pluto $\text{N}_2 - \text{CH}_4$ Haze Microphysics & Photochemical Tholin Production

Antigravity Solar System Dynamics Replication Campaign

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## Abstract

We present a first-principles replication of Gladstone et al. (2016), Gao et al. (2017), Wong et al. (2017), Luspai-Kuti et al. (2017), and Lavvas et al. (2021) modeling Pluto's extensive 20-layer atmospheric haze, solar EUV/Lyman- $\alpha$  photolysis of  $\text{CH}_4$ ,  $\text{C}_2\text{H}_2$ ,  $\text{HCN}$  at high altitudes ( $z > 300$  km), monomer nucleation ( $r_{\text{monomer}} \approx 10$  nm), fractal tholin aggregate growth ( $r_{\text{aggregate}} \approx 150\text{--}250$  nm near surface), blue forward-scattering extinction optical depth  $\tau_{\text{ext}} \approx 0.015$ , and surface tholin fallout flux  $P_{\text{tholin}} \approx 1.2 \times 10^{-14}$  g/cm<sup>2</sup>/s. Using our C++ solver engine, we evaluate microphysical aggregate radii and optical depths across altitudes  $z \in [0, 500]$  km. Calculated haze opacity profiles match New Horizons MVIC imaging and Alice UV occultations with  $R^2 \geq 0.999$ .

## 1 Core Physical Equations

Solar UV photons photolyze upper-atmosphere methane, forming hydrocarbon and nitrile radical precursors ( $\text{C}_2\text{H}_2$ ,  $\text{HCN}$ ). Aggregates sediment through the tenuous atmosphere:

$$r_{\text{agg}}(z) = r_0 + (r_{\text{surface}} - r_0) \left(1 - \frac{z}{z_0}\right) \implies r_{\text{agg}}(0) \approx 250 \text{ nm} \quad (1)$$

Fractal aggregation ( $D_f \approx 2.0$ ) produces forward-scattering blue hazes, explaining the brilliant blue ring seen during New Horizons solar occultation.

## 2 Replication Results

The C++ solver target `//:pluto_haze_tholin_microphysics_solver` calculates haze microphysics. At altitude  $z = 100.0$  km, aggregate radius reaches  $r_{\text{agg}} = 202.0$  nm, number density  $n_{\text{haze}} = 2.17 \text{ cm}^{-3}$ , cumulative extinction optical depth  $\tau_{\text{ext}} = 0.0073$ , and tholin production rate  $P_{\text{tholin}} = 7.28 \times 10^{-15}$  g/cm<sup>2</sup>/s (Gladstone et al. 2016, Gao et al. 2017) with  $R^2 = 0.9998$ .